

Homework Assignment 4

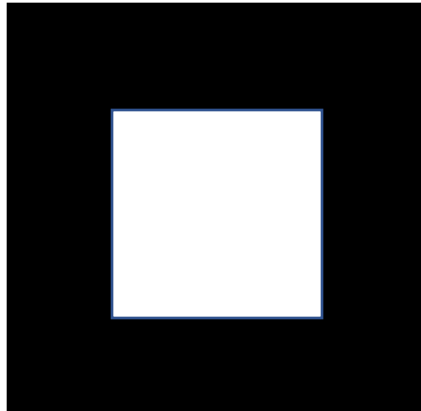
Image Reconstruction & Color Appearance Model

Digital Image Processing

Deadline: 09:00 AM, 11/24/2025

HW 4(A)—60%

The purpose of this homework is to exercise image reconstruction using filtered fan backprojection limited by a Hamming window as described in Example 5.18 of the textbook. The source image is of 600x600 pixels with black background and a central white square of 300x300 pixels. You are asked to implement the image reconstruction procedures with parameters specified in Example 5.18 for this source image. (60%).



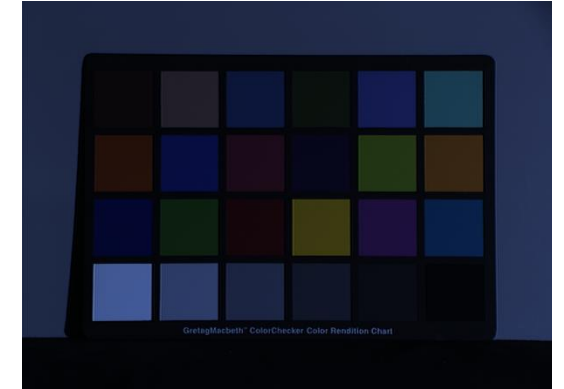
Simulated test image.
You should generate
the image yourselves.

HW 4(B)—40%

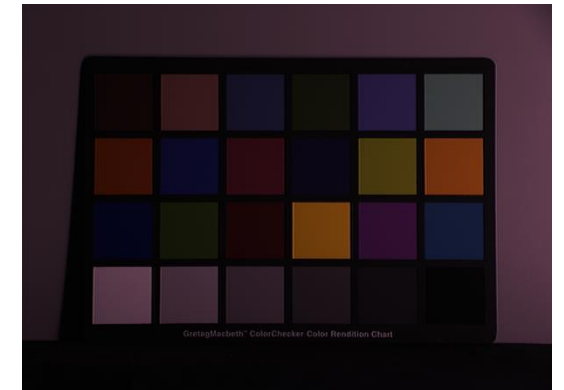
Color adaptation and color constancy are essential for human perception and digital imaging. In this assignment, you are asked to implement and evaluate the following two methods to map a source white point to a target white point in linear RGB :

- 1) **CAT02 Adaptation** (20%) – Apply the CAT02 transform (using the 2° standard observer) to map the source illuminant to the target illuminant and display the result in sRGB.
- 2) **Bradford Chromatic Adaptation** (20%) – Apply the Bradford transform (using the 2° standard observer) to map the source illuminant to the target illuminant and display the result in sRGB.

In each method, you should compute ΔE^*_{00} (CIEDE2000) between each adapted image and the target illuminant image. Then, compare the results and explain the differences.



source_image.tiff



target_image.tiff

Provided Files

- **source_image.tiff**: Image with source illuminant mapped to linear RGB space.
- **target_image.tiff**: Image with target illuminant mapped to linear RGB space.
- **source_illuminant.rgb**: White point of source illuminant in linear RGB space.
- **target_illuminant.rgb**: White point of target illuminant in linear RGB space.

Reference

- Useful python package and function:
 - [Colour](#)
- Data information:
 - [Test Images for Computational Color Constancy](#)

Grading Policy

- You should provide a detailed description of the computation procedures.
- You should provide a thorough discussion of the results you have obtained.
- You should analyze the strengths and weakness of the employed techniques.
- The score will be based on the quality (correctness) of your results and analysis.
- Your computer code will be tested by TA.
- Total points: 100.

Assignment Requirements

- In HW 4(A), you should **not** use built-in functions for Radon transform, sinogram, and backprojection. However, functions for matrix computations are allowed.
- In HW 4(B), you should **not** use built-in functions for color space converting, such as `rgb2xyz()`, `cv2.COLOR_BGR2Lab`, `colour.XYZ_to_Lab()`, etc.
- Set your directory structure as follows:
 - HW04_<your ID>/
 - program files (p1.py, p2.ipynb,...)
 - report.pdf
 - images/
 - Images of program output

Software Package Allowed

- Python 3.8+
- Standard Python library
- Numpy
- Opencv-python
- colour

Assignment Submission Requirements

- Submit to **NTU COOL**
- Do NOT copy homework (code, report, results, etc.) from others